Proprietary to General Electric Company
Direction 5438996-8EN, **Rev#** 9
Brivo CT385 Series Service Methods (Adv)
Module Version 1.0, **Version Date** 1/13/13

Plane of Rotation (POR) Alignment

(Click Here for Summary Version)

1 Personnel Requirements

Required Persons	Preliminary Reqs	Procedure	Finalization
1	Not Applicable	Not Applicable	Not Applicable

2 Overview

The purpose of Plane of Rotation measurement/alignment is to put the X-ray tube in the correct physical relationship to the detector and verify it. This is normally only necessary when the X-ray tube has been replaced or moved.

Illustration 1: POR Flowchart

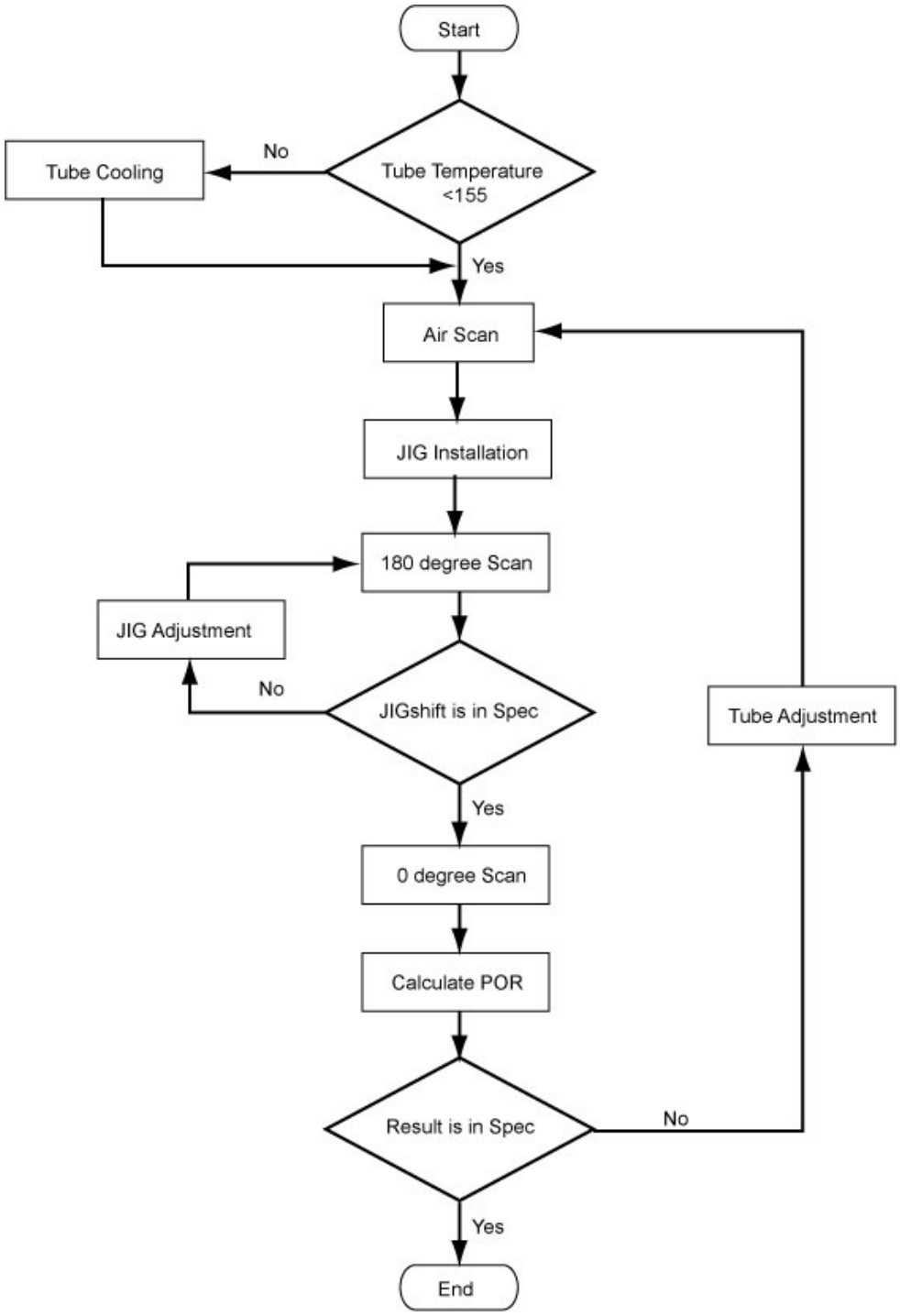


Illustration 2: POR JIG

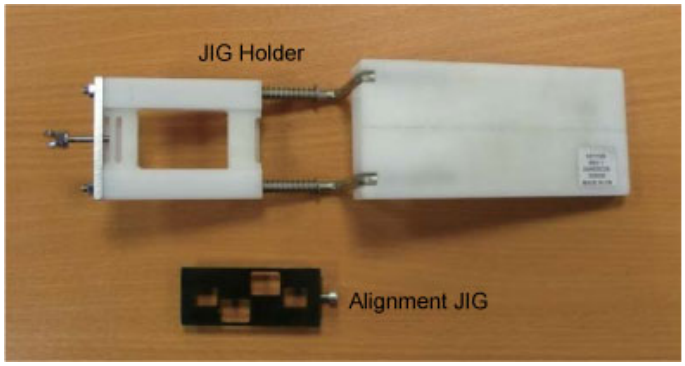
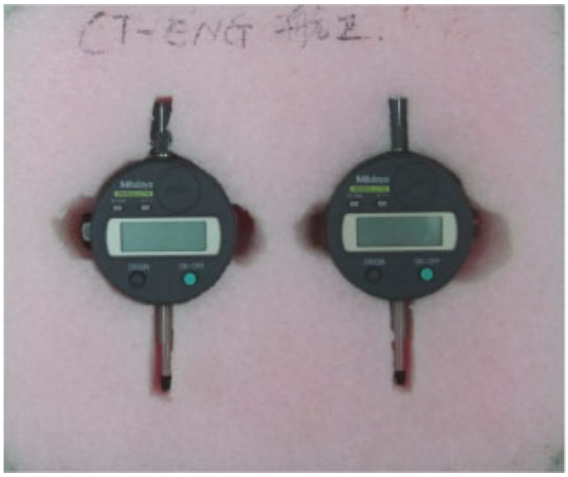


Illustration 3: Dial Gauge Indicator



3 Preliminary Requirements

3.1 Tools and Test Equipment

Item	Qty	Effectivity	Part#	Manufacturer
Alignment JIG Holder - see Illustration 2	1	-	5371128	-
Alignment JIG - see Illustration 2	1	-	5371129	-
Dial Gauge Indicator (millimeters or inches) - see Illustration 3	1	-	-	-

3.2 Safety

⚠ CAUTION

No personnel or patients should be in the area of the gantry while scanning without adequate radiation safety precautions being utilized.

3.3 Required Conditions

Condition	Reference	Effectivity
Recommend procedure completion by GE Service trained personnel.	-	-
The positioning lights must be OK before POR alignment.	-	-

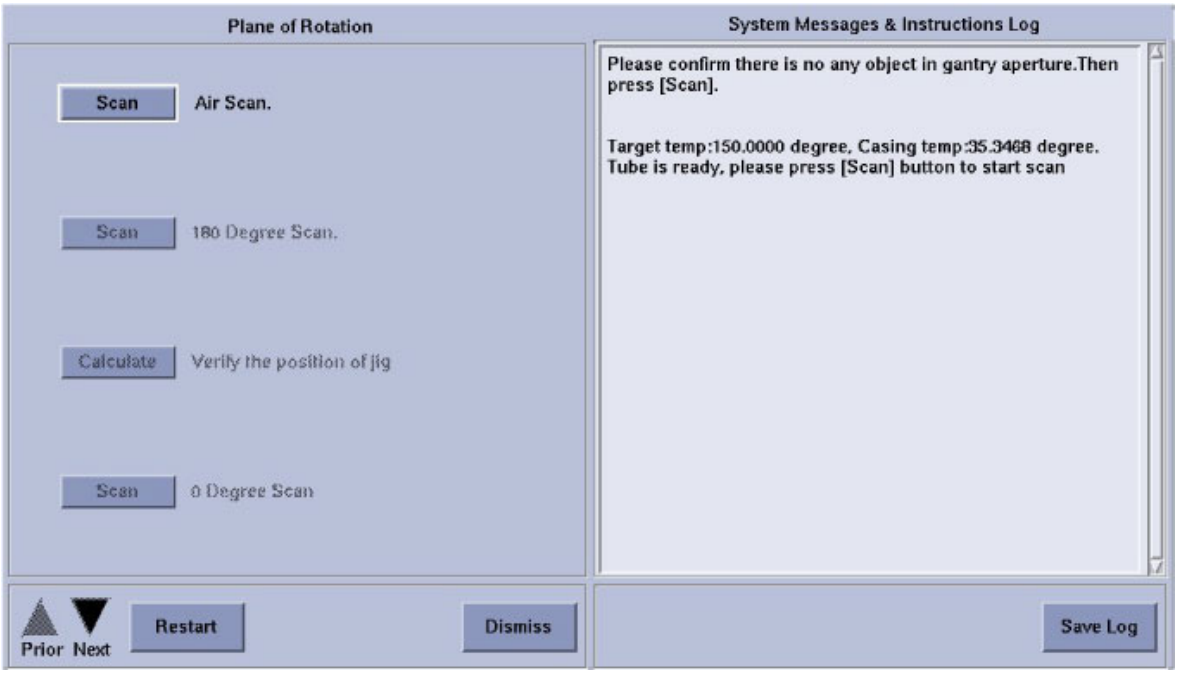
4 Procedure

4.1 Preparation

1. Move the cradle OUT to its fully retracted position (farthest from gantry) and position the table to its lowest height.
2. Remove the gantry left side cover.
3. Turn OFF “Axial Enable”, “HVDC” and “120VAC” switches on the service switch panel.
4. Remove the table foot switch cover.
5. Remove the gantry covers as required.
6. Turn ON “Axial Enable”, “HVDC” and “120VAC” switches.

4.2 JIG Installation and Adjustment

1. Select [Service] ->[Calibration] -> [POR Alignment] to enter 'Plane of Rotation' UI.
2. Click [Scan] to start **Air Scan**.

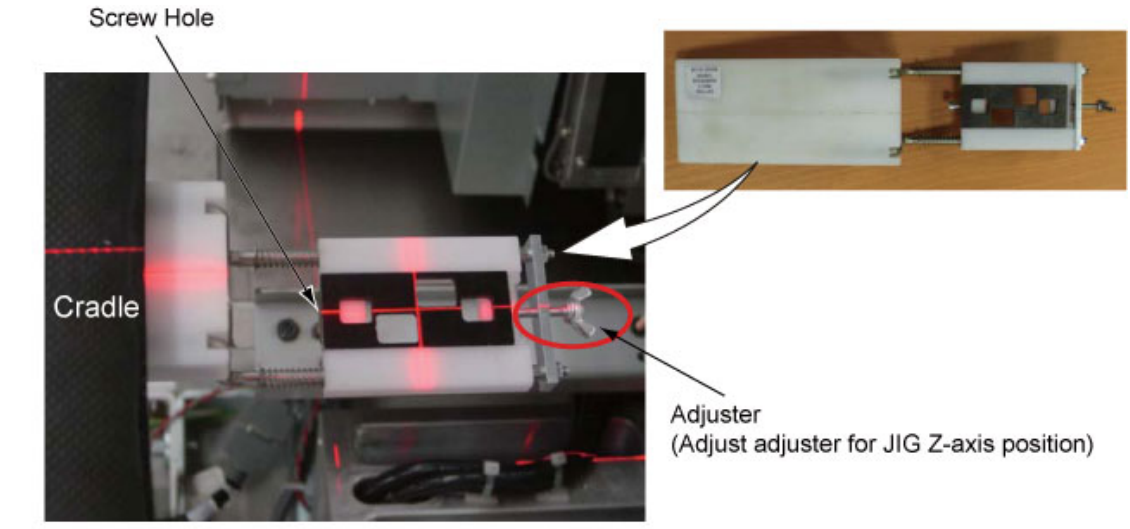


3. Hint information 'Air scan succeeded.' displays on the screen.
4. Install and adjust the POR JIG by referring information displayed on the screen.
   a. Elevate the table to 135mm height.
   b. Place the JIG onto the JIG holder and insert it into the cradle.

      Align the localization light along the two middle windows edge of tool as shown in Illustration 4.

      NOTE: Pay attention to the JIG direction on the JIG holder if screw is not on the JIG.

Illustration 4: POR JIG Installation



   c. Press [External Landmark] button to position the positioning light to the **190**.

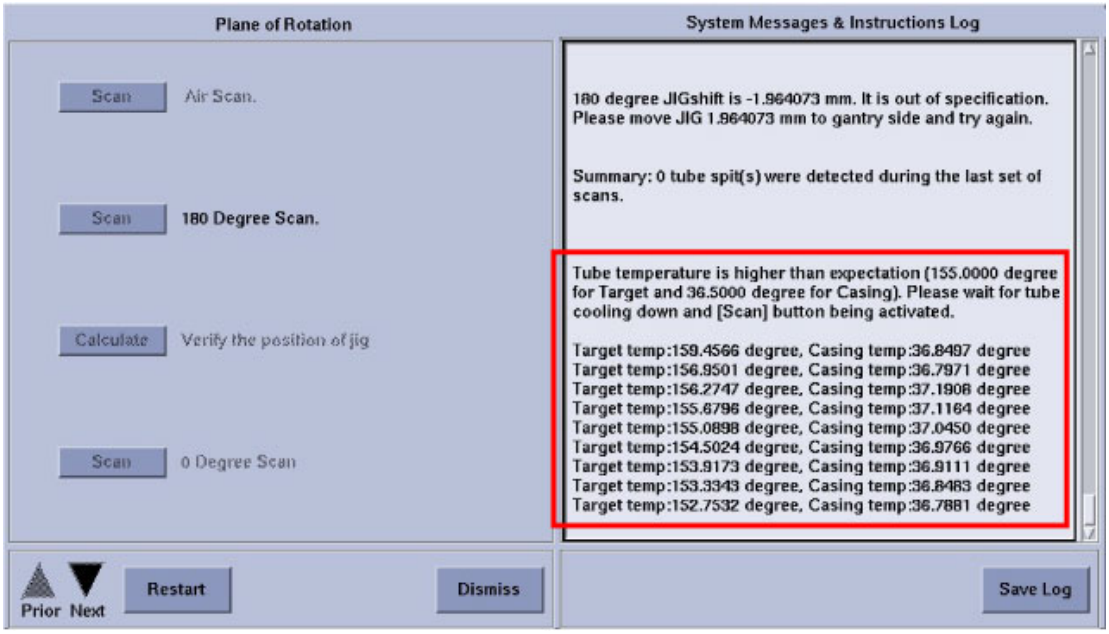
Illustration 5: External Landmark



d. Press [Cradle In] button to move the cradle until **0** displays on the display/control panel.

e. Click [Scan] to start **180 Degree Scan**.

NOTE: If the tube temperature is higher, the following hint information displays: `Tube temperature is higher than expectation. Please wait for tube cooling down and [Scan] button being activated.`



f. Hint information '180 degree jig scan succeeded.' displays on the screen.

g. Click [Calculate] to verify the position of JIG.

i. If it is in specification, the following information displays: `180 degree JIGshift is 0.xxxxxmm. It is in specification.`

ii. If it is out of specification, the similar information displays: `180 degree JIGshift is -1.970411mm. It is out of specification. Please move JIG 1.970411mm to gantry side and try again.`

Adjust the JIG position by using adjuster (See Illustration 4):

- If JIGshift is positive, turn adjuster CW, toward the table
- If JIGshift is negative, turn adjuster CCW, toward the gantry.

Repeat Step 4.e to Step 4.g until the calculate result is in specification.

5. Click [Scan] to start **0 Degree Scan**.

6. Click [Calculate] to start **Calculate the amount of tube movement**.

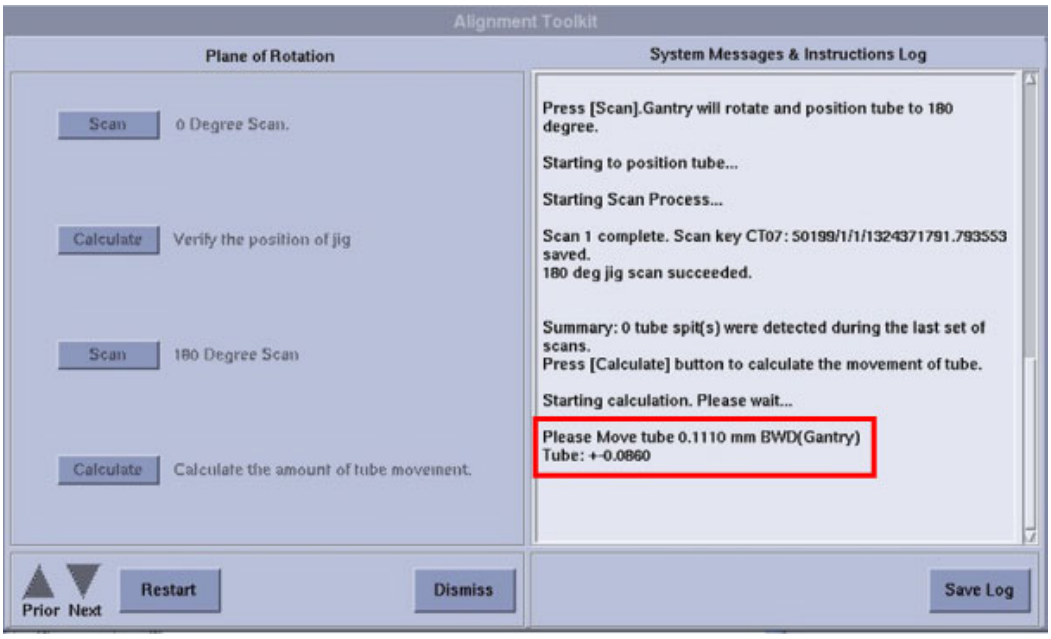
a. If it is in specification, the following information displays:

```
The shift of tube is xxxxx mm.

It is in specification: +/-0.0860 mm.
```

No need to adjust the position of the tube.

b. If it is out of specification, the following information displays as shown in figure:



7. Adjust the tube position.

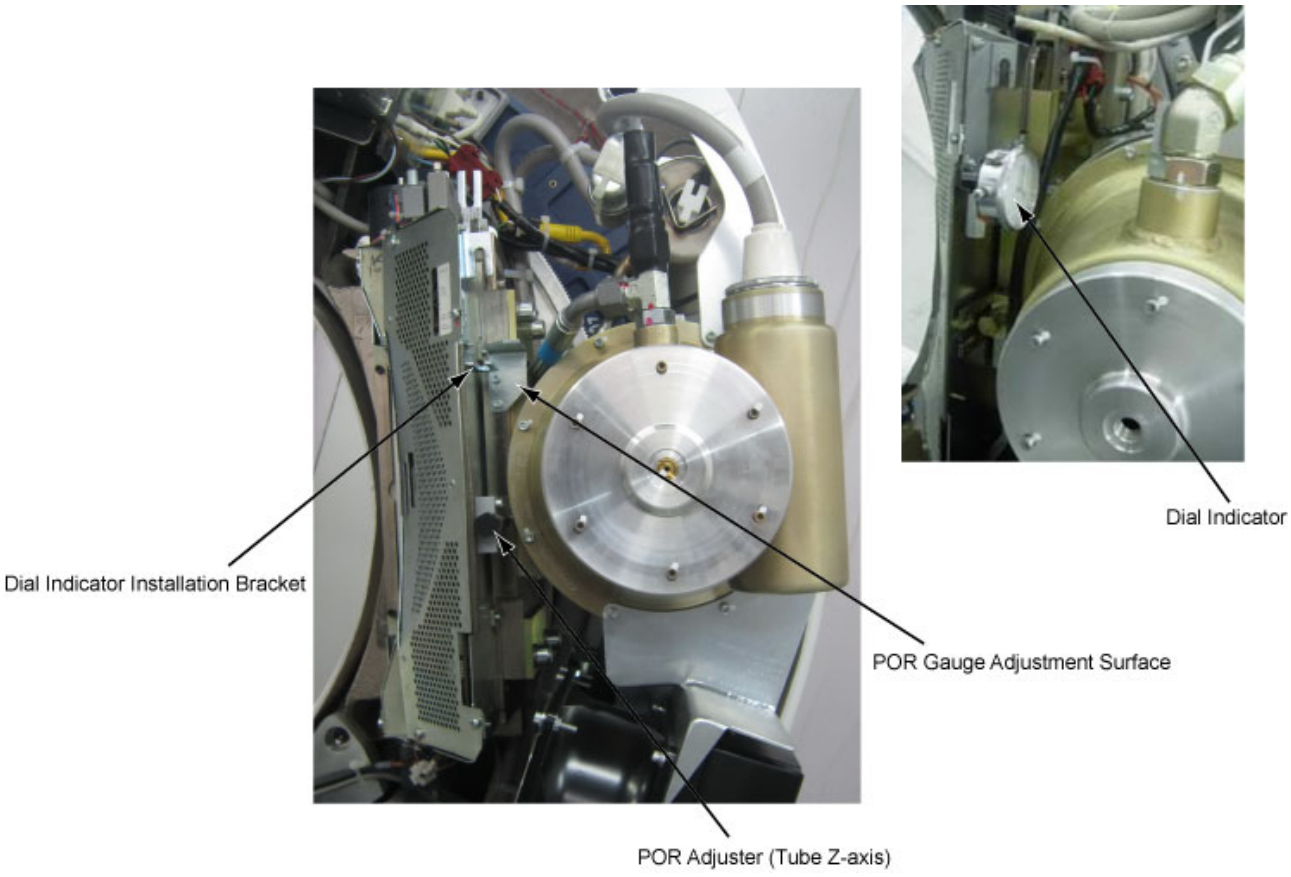
a. Rotate the gantry until the tube reaches about 3 o'clock position.

b. Turn OFF "HVDC", "Axial Enable" and "120VAC" switches on the service switch panel.

c. Mount the dial indicator onto the collimator / tube assembly as shown below. Make sure you zero the dial indicator, when it is securely in place.

NOTE: The dial indicator must be vertical with the POR gauge adjustment surface.

Illustration 6: Dial Indicator Installation



d. Loosen the four M12 mounting bolts that secure the tube. (1/2 turn out is all that's necessary, and not any more.)

Illustration 7: Tube Mounting Bolts



e. Adjust the tube position in Z-axis by turning the POR adjuster according to the calculative result displayed on the monitor screen.

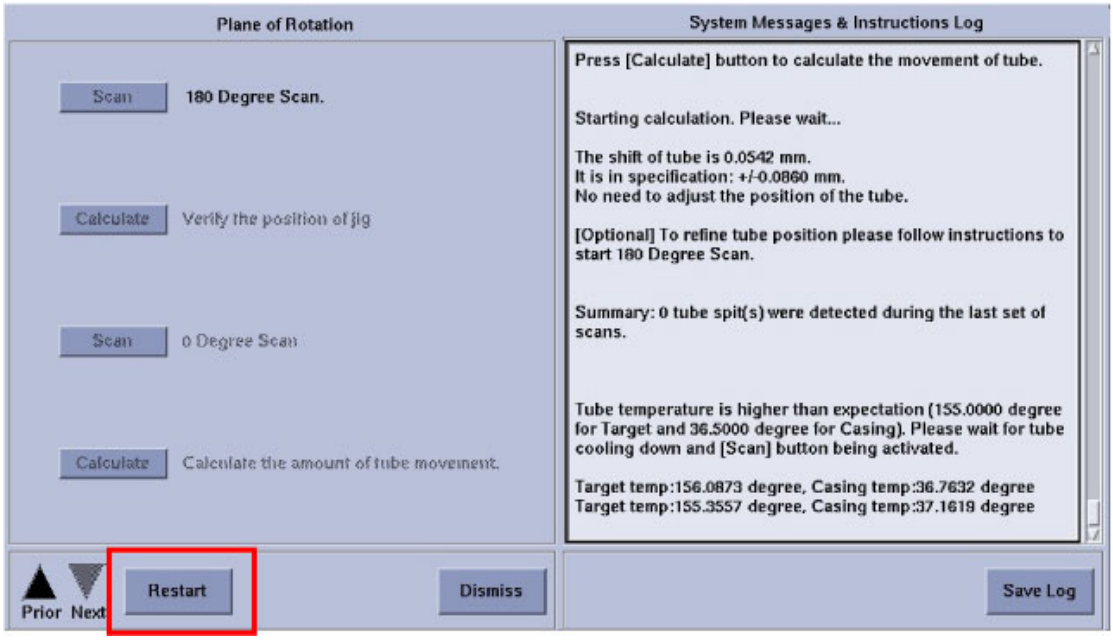
- If tubeshift is positive, turn adjuster CW
- If tubeshift is negative, turn adjuster CCW

8. Secure M12 mounting bolts to pre-load torque specification as shown below.

Table 1: Pre-load Torque Specification

25 ft.-lbs	33 N-m	295 in-lbs	340 kg-cm

9. Turn ON “Axial Enable”, “HVDC” and “120VAC” switches.
10. Click [Restart] to repeat POR procedure by referring to Step 2 to Step 6.



11. Click [Save Log], then click [Dismiss] to quit POR.
12. Apply final torque on all four M12 mounting bolts according to the values specified in Table 2.

Table 2: Final Torque

49 ft.-lbs	66 N-m	590 in-lbs	680 kg-cm

13. Restore the gantry to original configuration and reinstall the table foot switch cover.

5 Finalization

No finalization steps.

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